

1. The following concentrations were measured in a seawater sample collected from a station in the southern Pacific Ocean (so-called “Pacific and Indian Ocean Common Water”: depth = 3250 m; temperature = 1.6 C). The table also lists data for the concentrations found in pure “end-member” water masses that make up Common Water.

|                                                         | Salinity | [NO <sub>3</sub> ], (μmol / kg) | [O <sub>2</sub> ], (μmol / kg) |
|---------------------------------------------------------|----------|---------------------------------|--------------------------------|
| Pacific and Indian Ocean Common Water (measured sample) | 34.7     | 33                              | 180                            |
| North Atlantic Deep Water                               | 34.9     | 21                              | 260                            |
| Antarctic Intermediate Water                            | 34.3     | 33                              | 205                            |
| Antarctic Bottom Water                                  | 34.7     | 32                              | 240                            |

What are the relative proportions of the three water masses (NADW, AAIW, AABW) in the Common Water?

What assumptions do you need to make to use these data?

[Note that you can use linear algebra (matrices) to set up a system of equations and solve the unknowns – this can be much faster than the algebraic approach of solving for one variable in terms of the others, then plugging it in, etc.]

2. Two water masses mix, and each contributes 50%. The conventional  $^{14}\text{C}$  age of the two chemically identical components was determined to be 1000 years and 2000 years. What is the conventional  $^{14}\text{C}$  age of the resulting water mass?

3. a. You have a sediment core from a region of the ocean that has a constant sedimentation rate of  $1 \text{ cm kyr}^{-1}$ . Sketch the relationship between the measured age of the sediment and depth if there was (i) no mixing and (ii) a well-mixed upper layer.

b. Assume the top 8 cm of the sediments are well mixed by bioturbation. Draw a mass balance of  $^{14}\text{C}$  in the sediments using a 1 layer model of the mixed layer incorporating input by particles from above, sedimentation out, and radioactive decay. Use the model to predict the  $^{14}\text{C}$  age of the sediments after deposition.

4. Calcite shells of *Morozovella velascoensis* (planktonic foraminifera) from sediment cores in the low-latitudes across the Paleocene-Eocene Thermal Maximum (PETM, 56 Ma) have been analyzed for stable oxygen isotopes on a mass spectrometer. The mean  $\delta^{18}\text{O}$  values of multiple shells from before and during the event were reported as  $\delta^{18}\text{O}_c = -3.64\text{‰}$  and  $-4.51\text{‰}$ , respectively.

a. Assume that large ice sheets were absent during that time and that the ocean's  $\delta^{18}\text{O}$  was 1‰ different from today's value (0‰). Was the ocean isotopically lighter or heavier than today? Explain why.

b. Use the following paleotemperature equation to estimate temperatures before and during the event:

$$T(^{\circ}\text{C}) = 16.0 - 5.17(\delta_c - \delta_w) + 0.092(\delta_c - \delta_w)^2$$

where  $\delta_c$  and  $\delta_w$  are the  $\delta^{18}\text{O}$  of calcite and water from which the calcite formed, respectively.

c. Do your temperature estimates pertain to the atmosphere, surface ocean, deep ocean, or sediments? Explain why.